

## Trained



train



Large labelled  
training set

Environment-specific  
features

A

Standard



Cross-env.

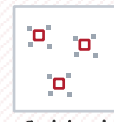


## Training-free, prior work



Frozen patch  
features

→



Codebook

or



Segmentation CLS attention

or



Prior-selected / aggregated features

B

Standard



Cross-env.



## Ours

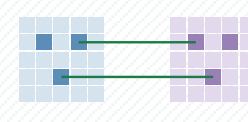


Frozen patch  
features

→



Within-image  
Balanced, matchable patch evidence



Across-image  
Balanced, matchable patch evidence

C

Standard

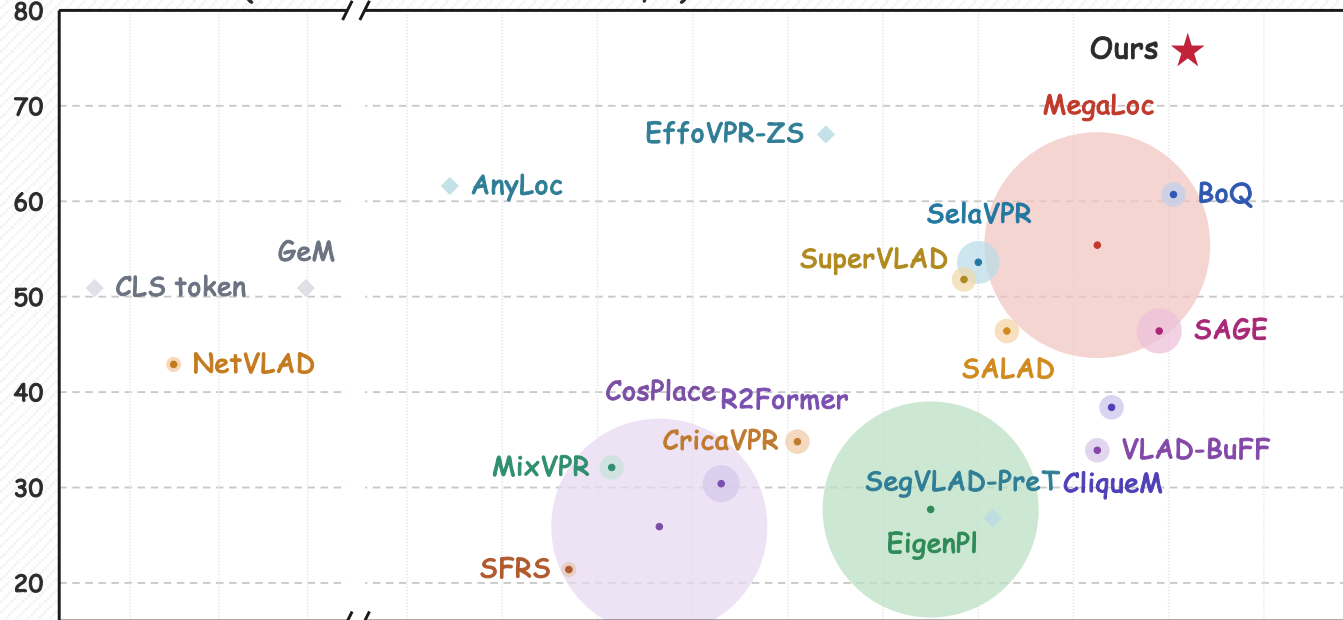


Cross-env.



## Training cost vs. performance

Laurel Caverns R@1 (cross-environment benchmark, %)



D

★ ours (training-free)

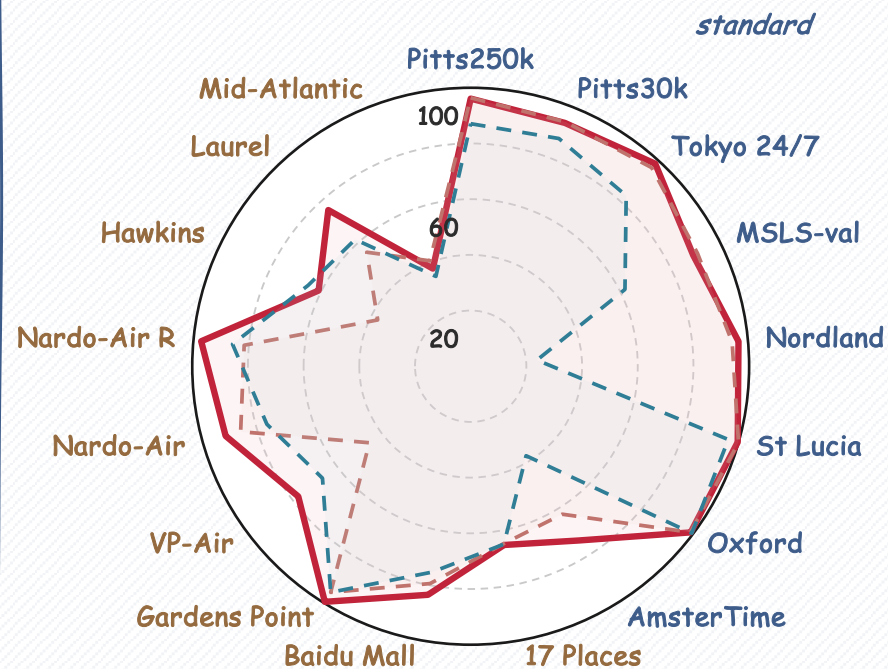
♦ training-free

● trained

● 1 million images

● 5 million images

## Performance on 17 benchmarks



cross-environment

E

— Ours

- - - MegaLoc

- - - AnyLoc